

DATA SCIENCE RESOURCES

Statistics for Data Science - Guide

Sure — I read the PDF and structured its content properly (clean headings + bullet format), so you can directly **copy-paste into Google Docs**.

Complete Learning Paths

Path 1: Absolute Beginner

1. Foundations

1. Descriptive statistics (mean, median, mode, variance)
2. Data visualization basics
3. Probability fundamentals
4. Practice with real datasets

Resources:

- Khan Academy Statistics: <https://www.khanacademy.org/math/statistics-probability>
 - StatQuest YouTube Playlist: <https://www.youtube.com/c/joshstarmer>
 - Think Stats (Free Book): <https://greenteapress.com/thinkstats2/html/index.html>
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2. Distributions & Inference

1. Normal, binomial, Poisson distributions
2. Central Limit Theorem
3. Confidence intervals
4. Hypothesis testing basics

Resources:

- Seeing Theory (Interactive): <https://seeing-theory.brown.edu/>
 - Crash Course Statistics: https://www.youtube.com/playlist?list=PL8dPuuaLjXtNM_Y-bUAhbISAdWRnmBUcr
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3. Practical Application

1. Correlation and regression
2. Python implementation (NumPy, Pandas, SciPy)
3. Real data analysis projects
4. Statistical visualizations

Resources:

- Python Statistics Tutorial: <https://realpython.com/python-statistics/>
 - Kaggle Learn Statistics: <https://www.kaggle.com/learn>
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Intermediate

Focus Areas

1. Advanced hypothesis testing (t-tests, ANOVA, chi-square)
2. Multiple regression and regularization
3. Logistic regression
4. Bayesian statistics intro
5. A/B testing and experimental design
6. Time series basics

Resources:

- Penn State Online Statistics: <https://online.stat.psu.edu/statprogram/>
- StatQuest Advanced Topics: <https://www.youtube.com/c/joshstarmer>
- Bayesian Methods for Hackers:
<https://github.com/CamDavidsonPilon/Probabilistic-Programming-and-Bayesian-Methods-for-Hackers>

Advanced

Focus Areas

1. Multivariate statistics and PCA
2. Advanced Bayesian methods
3. Causal inference
4. Time series forecasting (ARIMA, Prophet)
5. Survival analysis
6. Monte Carlo methods
7. Bootstrapping and resampling

Resources:

- Statistical Rethinking (Book): <https://xcelab.net/rm/statistical-rethinking/>
 - Elements of Statistical Learning (Free): <https://hastie.su.domains/ElemStatLearn/>
 - Coursera – Bayesian Statistics: <https://www.coursera.org/specializations/bayesian-statistics>
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Python Libraries for Statistics

Essential Libraries

1. NumPy – Numerical Computing

Documentation: <https://numpy.org/doc/stable/reference/routines.statistics.html>

2. Pandas – Data Manipulation and Analysis

Documentation: <https://pandas.pydata.org/docs/reference/frame.html>

3. SciPy – Scientific Computing and Stats

Documentation: <https://docs.scipy.org/doc/scipy/reference/stats.html>

4. Statsmodels – Statistical Modeling

Documentation: <https://www.statsmodels.org/stable/index.html>

5. Matplotlib & Seaborn – Visualization

Seaborn Tutorial: <https://seaborn.pydata.org/tutorial.html>

Statistics Concepts by ML Algorithm

Understanding what statistics powers what ML

Linear Regression → Statistics

- Ordinary Least Squares
 - R^2 and adjusted R^2
 - Residual analysis
 - Assumption checking
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Logistic Regression → Statistics

- Maximum Likelihood Estimation
 - Odds ratios
 - Log-odds
 - ROC curves
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Naive Bayes → Statistics

- Bayes' Theorem
 - Conditional probability
 - Prior and posterior distributions
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Decision Trees → Statistics

- Information gain (entropy)
 - Gini impurity
 - Chi-square test for splits
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Neural Networks → Statistics

- Gradient descent (optimization)
 - Maximum Likelihood
 - Regularization (L1/L2)
 - Initialization (normal distribution)
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K-Means Clustering → Statistics

- Euclidean distance
 - Variance minimization
 - Within-cluster sum of squares
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PCA → Statistics

- Covariance matrix
- Eigenvalues and eigenvectors
- Variance explained

☀ Next Steps After This Guide

Where to Go From Here

1. Build a Strong Foundation

- Master all concepts in sections 1–6
 - Can take 3–6 months
 - Don't rush, depth > breadth
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2. Get Practical

- Start coding in Python/R
 - Complete 5–10 small projects
 - Share on GitHub
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3. Specialize Based on Interest

- Business Analytics: A/B testing, causal inference
 - Machine Learning: Bayesian methods, regularization
 - Research: Advanced inference, experimental design
 - Finance: Time series, risk modeling
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4. Join Communities

- r/statistics: <https://www.reddit.com/r/statistics/>
- Cross Validated: <https://stats.stackexchange.com/>
- Kaggle Forums: <https://www.kaggle.com/discussion>
- LinkedIn groups and Discord servers

5. Keep Learning

- Follow stat blogs and podcasts
 - Read papers in your field
 - Take advanced courses
 - Contribute to open source
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6. Build Portfolio

- Document learning journey
 - Write technical blog posts
 - Create tutorials
 - Help others learn
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30-Day Statistics Challenge

Day-by-Day Learning Plan

Week 1: Descriptive Statistics

- Day 1–2: Mean, median, mode, range
 - Day 3–4: Variance, standard deviation
 - Day 5–6: Quartiles, IQR, boxplots
 - Day 7: Practice problems + Python implementation
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Week 2: Probability & Distributions

- Day 8–9: Basic probability rules
- Day 10–11: Conditional probability, Bayes' theorem

- Day 12–13: Normal distribution
 - Day 14: Binomial and Poisson distributions
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Week 3: Inference & Testing

- Day 15–16: Sampling and CLT
 - Day 17–18: Confidence intervals
 - Day 19–20: Hypothesis testing fundamentals
 - Day 21: t-tests and p-values
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Week 4: Relationships & Application

- Day 22–23: Correlation
 - Day 24–25: Simple linear regression
 - Day 26–27: Multiple regression
 - Day 28–29: Logistic regression
 - Day 30: Final project combining all concepts
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Final Note

The goal isn't to become a statistician, it's to become a data professional who uses statistics as a powerful tool for understanding the world.

Additional Resources

For Visual Learners

- 3Blue1Brown: <https://www.youtube.com/c/3blue1brown>
- StatQuest: <https://www.youtube.com/c/joshstarmer>

- Seeing Theory: <https://seeing-theory.brown.edu/>
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For Hands-on Learners

- Kaggle Learn: <https://www.kaggle.com/learn>
 - DataCamp: <https://www.datacamp.com/>
 - Real Python: <https://realpython.com/>
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Good luck on your statistics journey! Remember: Every expert was once a beginner. Stay consistent, stay curious, and keep practicing.
